

AI POWERED MOCK DATA GENERATOR AGENT

ABSTRACT

The AI Powered Mock Data Generator Agent is a web-based application developed to automate the generation of realistic mock datasets for software development and testing. The application accepts SQL DDL or YAML schemas as input, analyzes the structure, and generates synthetic data while respecting schema constraints. The generated data can be exported as CSV files and SQL INSERT statements. The system also integrates external APIs and follows an Agent Loop architecture for intelligent data processing.

INTRODUCTION

Software developers and QA engineers require large amounts of sample data for testing applications. Creating realistic datasets manually is time-consuming and often fails to satisfy testing requirements. The AI Powered Mock Data Generator Agent solves this problem by automatically generating realistic test data based on user-defined schemas.

OBJECTIVE

- To generate realistic mock datasets automatically.
- To support SQL DDL and YAML schema formats.
- To reduce manual effort in software testing.
- To export generated data as CSV and SQL files.
- To implement an AI Agent Loop for intelligent processing.

PROBLEM STATEMENT

➤ Software developers and QA engineers require large amounts of realistic test data for application testing and database validation. Creating such data manually is time-consuming, error-prone, and often does not satisfy schema constraints or real-world patterns. Moreover, using actual user data for testing may lead to privacy and security concerns. Therefore, there is a need for an automated system that can generate realistic mock data from SQL DDL or YAML schemas while maintaining data consistency and supporting easy export for testing purposes.

TECHNOLOGIES USED

- Python
- Streamlit
- Faker Library
- Pandas
- PyYAML
- Matplotlib
- REST Countries API
- Pytest
- GitHub

SYSTEM ARCHITECTURE

User Input



Schema Parsing



Agent Loop Engine



Data Type Mapping



Constraint Validation



Mock Data Generation



CSV / SQL Export



Dashboard Display

MODULES

Schema Parser Module

- Reads SQL DDL and YAML schemas.
- Identifies column names and data types.

Agent Loop Module

- Parses schema.
- Maps data types.
- Generates realistic values.
- Validates generated data.
- Exports results.

Mock Data Generator Module

- Generates names, emails, phone numbers, cities, salaries, departments, and other realistic data.
- Uses the Faker library for data creation.

Dashboard Module

- Displays generated data.
- Shows live agent logs.
- Provides CSV and SQL download options.

Testing Module

- Executes automated test cases.
- Validates application functionality using Pytest.

Workflow

Schema Input



Parse



Map



Generate



Validate



Export



Display Results

AI Capabilities

The application follows a five-step AI Agent Loop:

Step 1: Parse Schema

Step 2: Map Data Types

Step 3: Generate Realistic Data

Step 4: Validate Constraints

Step 5: Export Results

Advantages

- Reduces manual effort.
- Generates realistic test data.
- Supports SQL and YAML schemas.
- Provides real-time processing logs.
- Integrates external APIs.
- Improves software testing efficiency.
- Open-source and cost-effective.

Hardware Requirements

Processor : Intel i3 or above

RAM : 4 GB Minimum

Storage : 500 MB

Operating System : Windows 10 / 11

Software Requirements

Python 3.9+

Visual Studio Code

Streamlit

Faker

Pandas

PyYAML

Matplotlib

Pytest

Git

Future Enhancements

- Database integration.
- User authentication.
- Cloud deployment.
- AI-powered smart schema suggestions.
- PDF and Excel export.
- Multi-user support.

Applications

- Software Testing
- Database Validation
- Data Engineering
- API Testing
- Educational Projects
- QA Automation
- Prototype Development

Conclusion

The AI Powered Mock Data Generator Agent is an intelligent solution for generating realistic mock datasets automatically. The project combines AI Agent Loop concepts, schema parsing, external API integration, and data visualization to simplify software testing. It improves productivity, reduces development effort, and provides an efficient platform for generating and managing synthetic data.